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Long Story Short

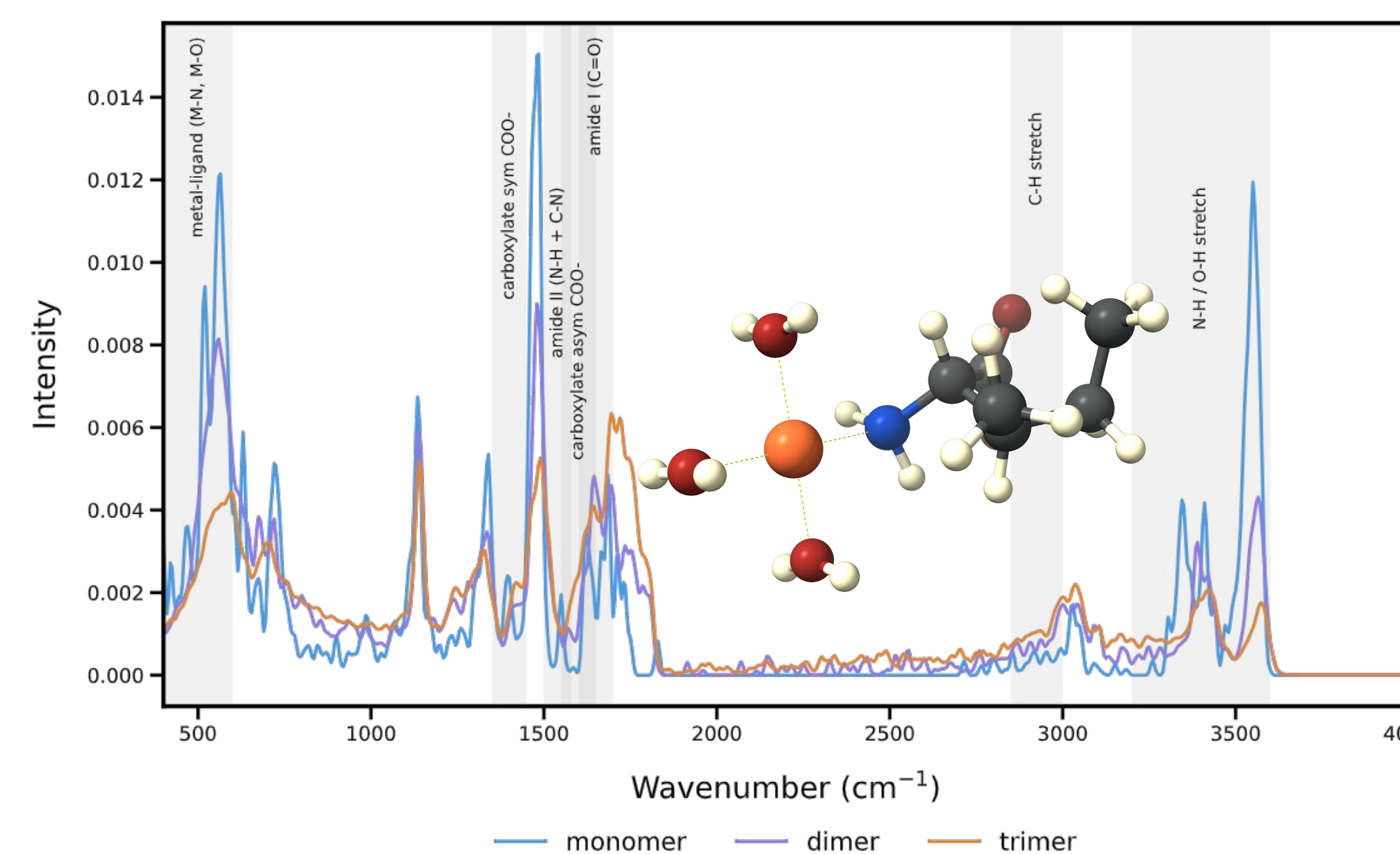
Metal identification that transfers from short peptides to long

Abstract

Transition metals sit unused in electronic and industrial waste, and peptides offer a practical way to reclaim them, since they can be tuned to bind and separate specific metal ions.¹ The catch is that you have to know which metal a given peptide has bound, and infrared spectroscopy offers an inexpensive way to read it, since every complex leaves a distinct vibrational fingerprint. Semiempirical quantum chemistry with xTB lets us generate those spectra in bulk before doing the experiments.² Common agents used for separation are longer peptides, which are more expensive to simulate and measure. So we test whether a model trained only on monomer metal-peptide complexes (monomers) and dimer metal-peptide complexes (dimers) can extend to longer metal-peptide complexes (trimers) not in the training set. Across 24,685 simulated mono/di IR spectra of Co, Ni, Cu, and Zn complexes, a random forest model³ classifies the metal at 0.81 macro-F1 per conformer on the mono/di test set, and transfers to the trimer test set at 0.63. In-domain, the model classifies all four metals well, with Ni and Co the hardest pair. Adding distinct trimers to the training set closes more of the gap than adding data volume alone. Given enough added trimers, a model anchored on short peptides reached the same plateau as one anchored on trimers. This points to a cost-efficient way to build such models, where we anchor on short metal-peptide complexes and add a modest, diverse set of longer ones to get a working model.

Data Visualization

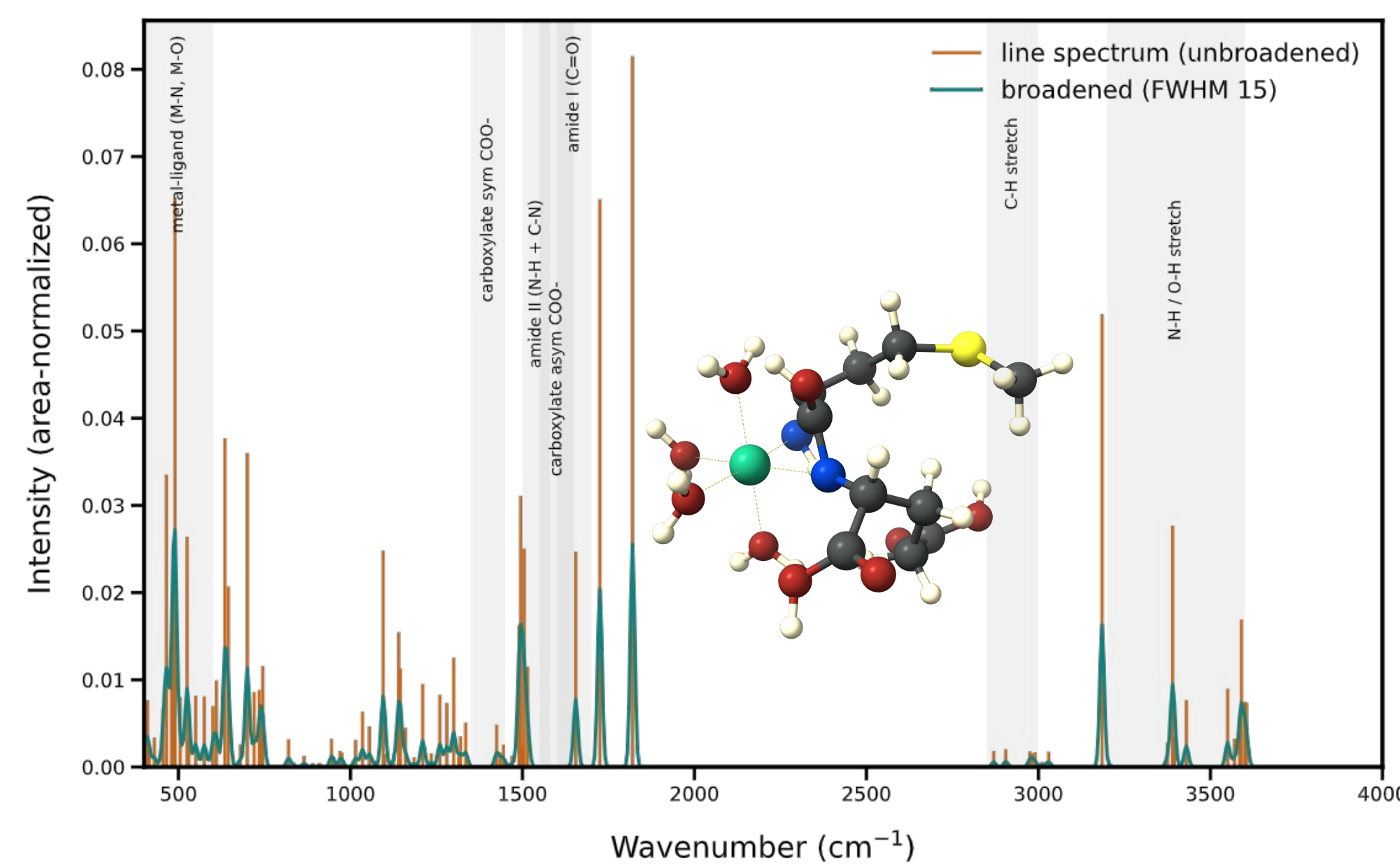
Simulated Cu²⁺ IR Spectra by Peptide Length



We simulated 128,854 IR spectra of Co, Ni, Cu, and Zn complexes on monomer, dimer, and trimer peptides (24,685 mono/di; 104,169 trimer), 791 intensity values each from 50 to 4000 cm⁻¹, binned per 5 cm⁻¹.

Data Featurization

Line Spectrum vs Gaussian Broadening

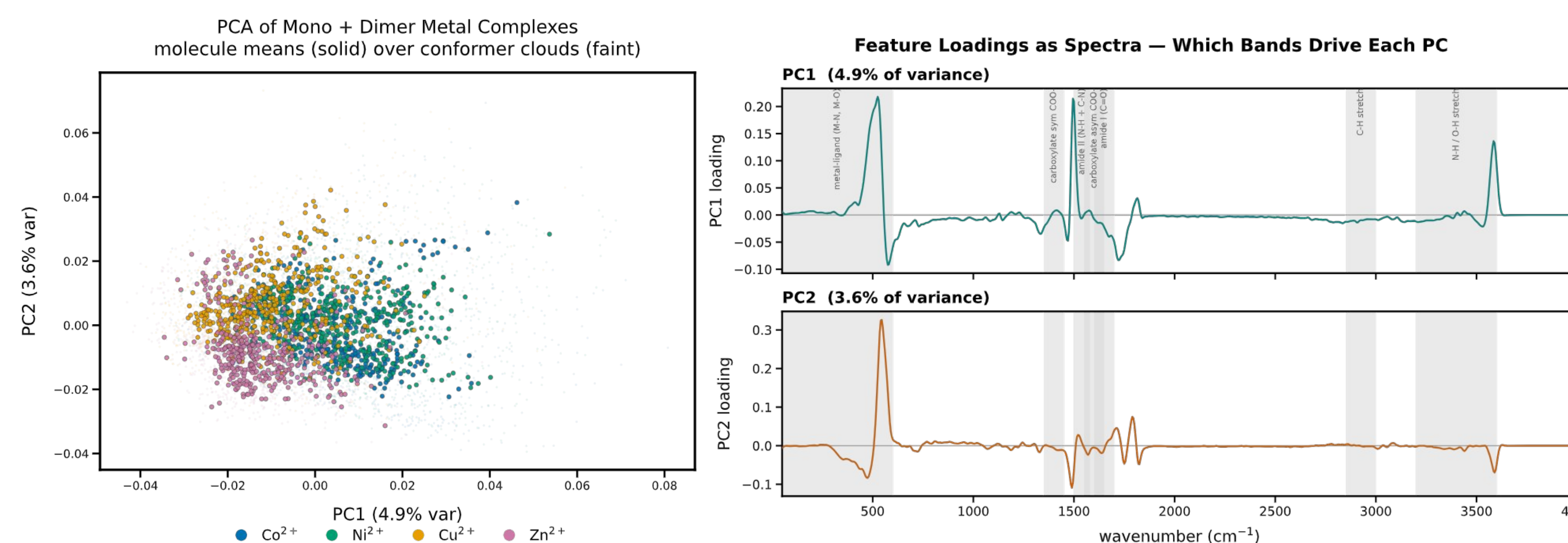


We applied Gaussian broadening to the spectra to make them more comparable and more like realistic, experimental spectra. With comparable features in hand, the next question is whether the metals separate easily.

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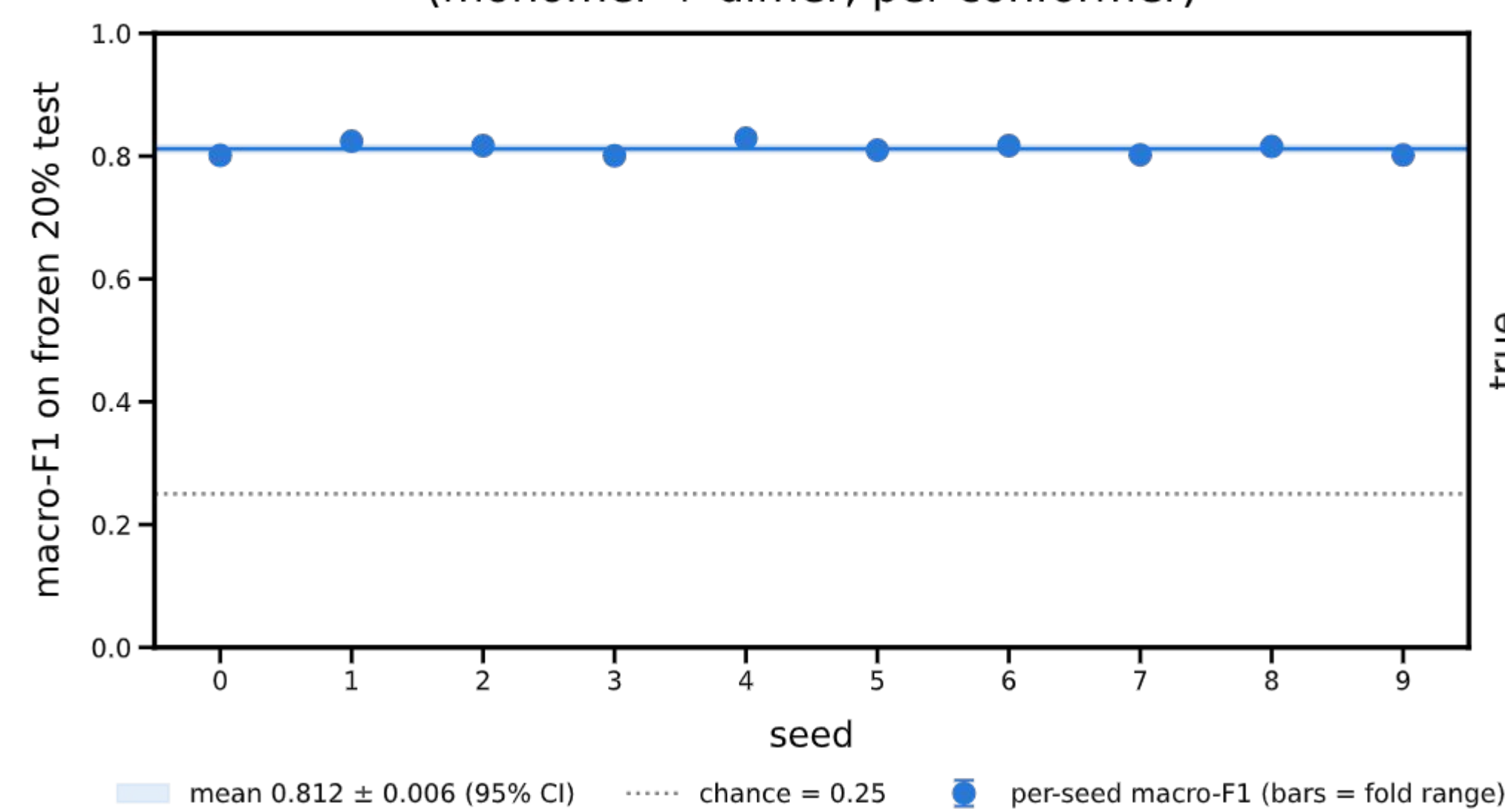
Data Analysis



We used principal component analysis (PCA) for exploratory data analysis to see whether natural grouping for the four metal systems exists in our data. However, the first two principal components capture only a total variance of 8.5%, which shows that the directions of greatest variance do not seem to separate the metal systems well. We do see some rudimentary separation between the four metals, but they do not seem to be linearly separable. So we chose to use a random forest model in future experiments.

Model Training

Metal classification — 10-seed Monte-Carlo CV (monomer + dimer, per-conformer)



Per-conformer model — per-metal confusion (out-of-fold, mean of 10 seeds)

	Co ²⁺	Ni ²⁺	Cu ²⁺	Zn ²⁺
Co ²⁺	0.80	0.15	0.03	0.02
Ni ²⁺	0.20	0.71	0.04	0.05
Cu ²⁺	0.04	0.03	0.89	0.04
Zn ²⁺	0.03	0.04	0.04	0.89

We needed a baseline model to see whether it classifies the metals on mono/di correctly. So we used a train/test split repeated over 10 random seeds, where each seed is just a different random split, so the score reflects 10 splits and not one lucky draw. The split is grouped by molecule, so all conformers of a molecule stay on one side. The baseline reaches about 0.81 macro-F1 per conformer. Notably, Ni and Co are the most confused pair. Co and Ni bind the peptide in nearly the same shape and have similar molar masses, so their spectra overlap; Cu and Zn each sit differently and stay easy to tell apart.

Zero-Shot Test on Trimers

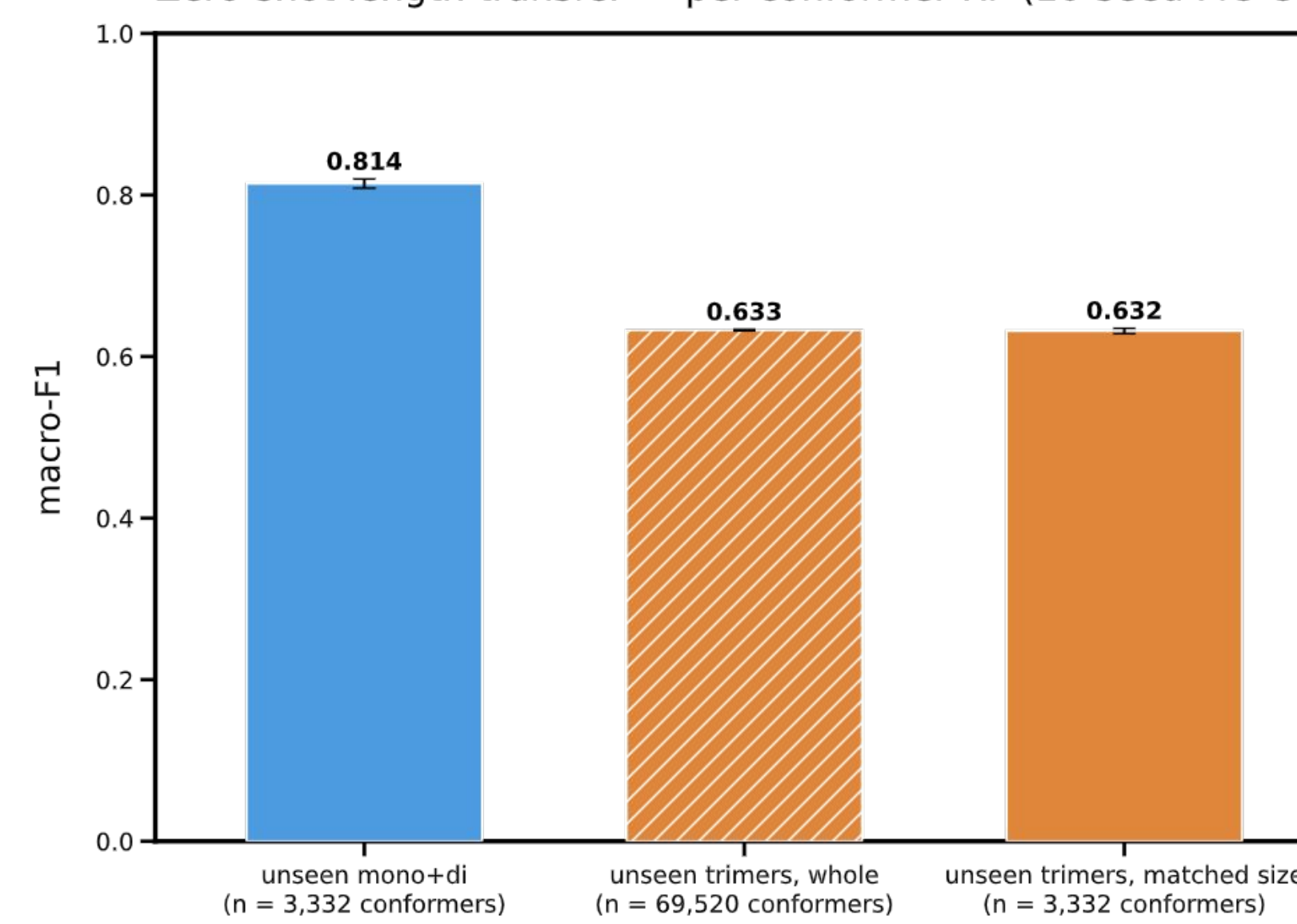
Mono+di model, zero-shot on all (whole) trimers (per-conformer, per-metal confusion, mean of 10 seeds)

	Co ²⁺	Ni ²⁺	Cu ²⁺	Zn ²⁺
Co ²⁺	0.61	0.25	0.08	0.06
Ni ²⁺	0.21	0.49	0.13	0.17
Cu ²⁺	0.07	0.07	0.69	0.17
Zn ²⁺	0.06	0.07	0.11	0.76

Mono+di model, zero-shot on matched-size trimers (per-metal confusion, mean of 10 seeds)

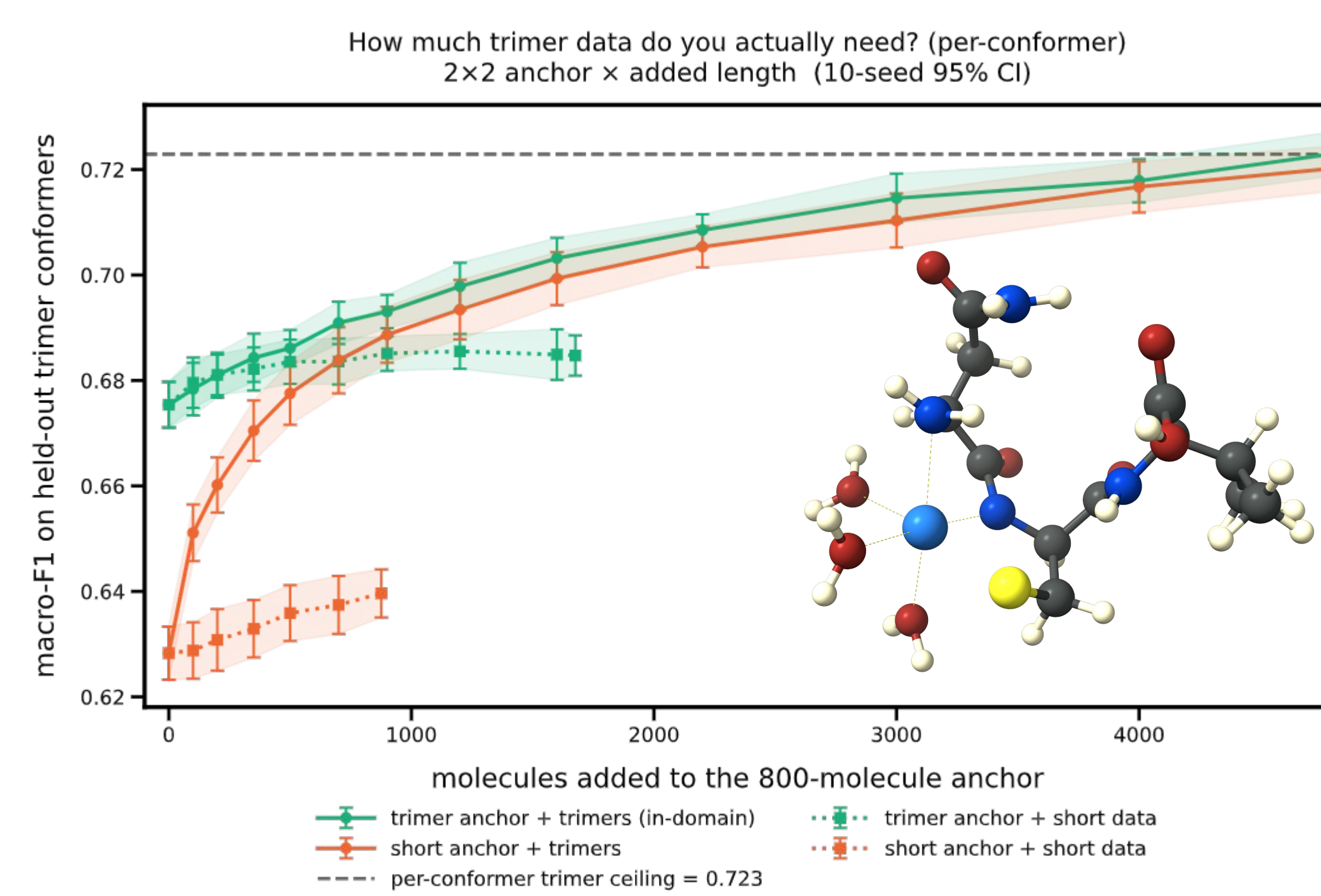
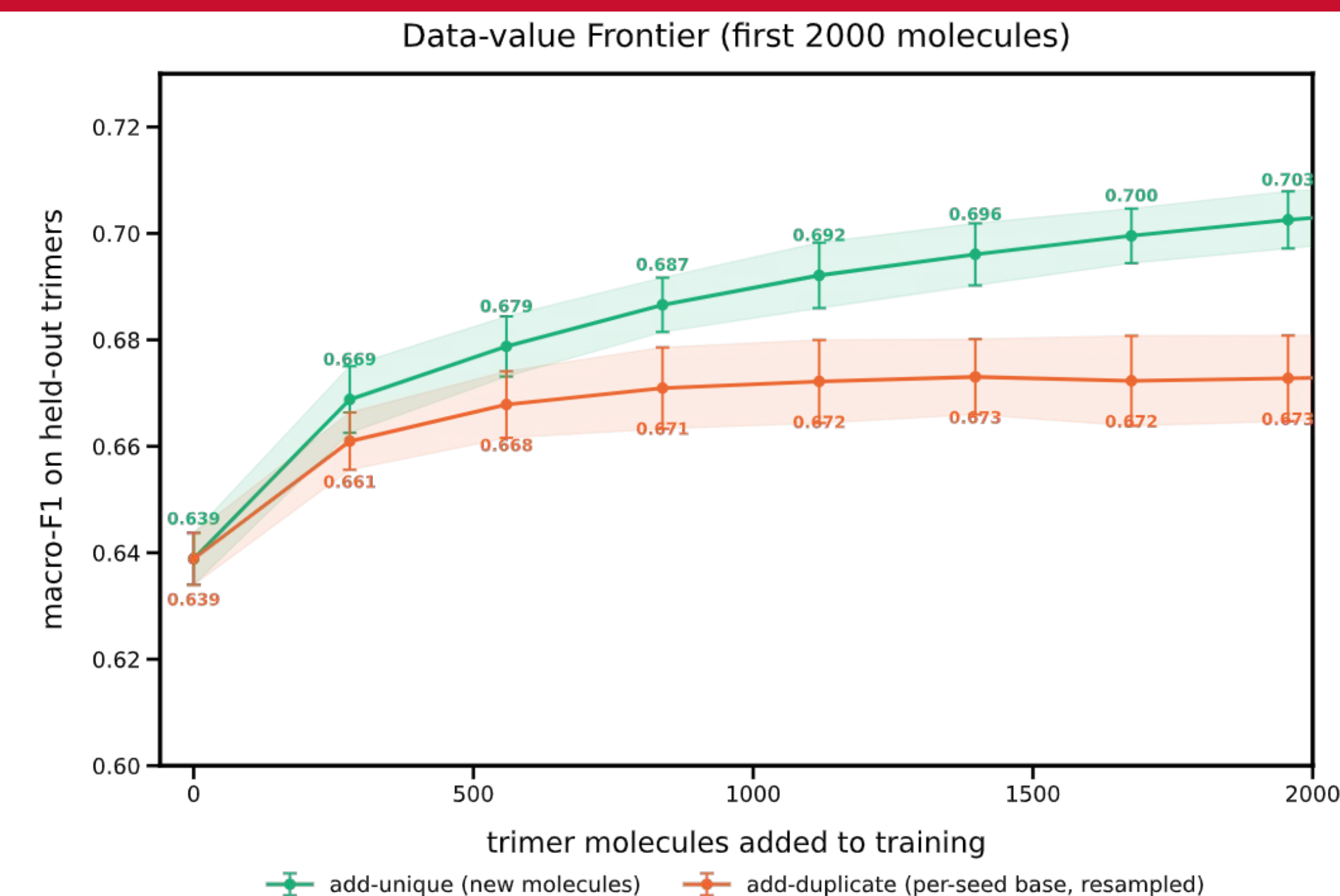
	Co ²⁺	Ni ²⁺	Cu ²⁺	Zn ²⁺
Co ²⁺	0.61	0.25	0.07	0.06
Ni ²⁺	0.22	0.49	0.12	0.17
Cu ²⁺	0.07	0.07	0.68	0.17
Zn ²⁺	0.05	0.08	0.12	0.75

Zero-shot length transfer — per-conformer RF (10-seed MC-CV)



Can a higher-level pattern be learned from its individual components? We trained a model only on monomers and dimers and used it to predict the bound metal of unseen trimers. We tested the mono/di model on a held-out mono/di set, a held-out trimer set, and a held-out trimer set the same size as the mono/di set. To rule out chance we used 10 seeds, and that same-size trimer set makes sure the drop isn't just from a bigger test. We found that the metal signal transfers, but not all the way. The model still names the metal in most trimers, just less reliably than in short peptides. Similar to our baseline, Ni carries over the worst, and Co and Ni blur together because they look almost the same to the model.

Adding Trimers to Training Set



The benefit of additional training data depends on its composition, not its volume. We grew the mono/di training set two ways. Adding distinct trimers raised macro-F1 toward the trimer in-domain ceiling, whereas duplicating a small trimer set plateaued below it. When testing on a held-out trimer set, given enough added trimers, a model anchored on short peptides reached the same plateau, near 0.73, as one anchored on trimers. Adding short peptides in place of trimers gave no gain at either anchor. The practical recipe follows: build on the abundant short peptides (mono/di), then direct new simulation toward distinct trimers, the only additions that raise trimer accuracy.

Conclusion

We trained a random forest model to identify the bound metal from a simulated IR spectrum, and a model built only on short peptides still identified the metal in unseen trimers, though less reliably. Adding distinct trimers to training recovered most of that loss, while duplicated data and extra short peptides did not, so the limit is the diversity of longer peptides rather than the amount of data. To cut the error, we will target the Co versus Ni confusion directly, testing spectral regions that separate their near-identical binding geometries and stronger classifiers such as gradient boosting or a convolutional network. We will also extend the length jump past trimers to tetramers and pentamers, and check the model against measured infrared spectra rather than simulated ones.

References

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